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[file,path]=uigetfile('*.','Select an image file');
img=imread(fullfile(path, file));

F=fft2(double(img));
magnitude = abs(F);
phase = angle(F);

reconstructed = ifft2(magnitude.*exp(1i*phase));
reconstructed = uint8(real(reconstructed));

subplot(1,2,1);
imshow(img);
title('Org. Image');
subplot(1, 2, 2);
imshow(reconstructed);
title('Recon. Image');

```

Org. Image



Recon. Image



```

Fs = 1000;
t=0:1/Fs:1-1/Fs;
f1=10;
x=cos(2*pi*f1*t);

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```

analytic_signal = hilbert(x);
phase = unwrap(angle(analytic_signal));
instantaneous_frequency = diff(phase)*Fs/(2*pi);

subplot(2,1,1);
plot(t, x);
title('original Signal');
xlabel('Time');
ylabel('Amplitude');

subplot(2, 1, 2);
plot(t(1:end-1), instantaneous_frequency);
title('Instantaneous Frequency');
xlabel('Time');
ylabel('Frequency(Hz)');

```

